TrackDancer — Two-Board Build Sheet v2 (distributed nodes)

SUPERSEDES the single-board v1. Board A rides with the Pi (dash box); Board B is the DAQ NODE and can mount anywhere — only the CAN pair (+ power) connects them. On the bench they sit side by side and the Teensy keeps its USB to the Pi. Check each box as you land it. POWERED OFF while soldering.

BOARD A — dash-side CAN interface (small: MCP2515 transplant + SN65HVD230 #2)

☐	**1.**	Pi 3V3 (pin 1)	→	A: 3V3 rail	*via cobbler ribbon or direct cores*
☐	**2.**	Pi GND (pins 6 + 14)	→	A: GND rail	*two cores*
☐	**3.**	Pi MOSI (pin 19)	→	MCP2515 SI	
☐	**4.**	Pi MISO (pin 21)	→	MCP2515 SO	
☐	**5.**	Pi SCLK (pin 23)	→	MCP2515 SCK	
☐	**6.**	Pi CE0 (pin 24)	→	MCP2515 CS	
☐	**7.**	Pi GPIO25 (pin 22)	→	MCP2515 INT	
☐	**8.**	Pi SDA / SCL (pins 3/5)	→	IMU SDA / SCL	*IMU lives on Board A, flat+rigid*
☐	**9.**	MCP2515 VCC + #2 3.3V	→	A: 3V3 rail	
☐	**10.**	MCP2515 GND + #2 GND	→	A: GND rail	
☐	**11.**	TXCAN tack wire (pad 1)	→	#2 CAN TX	*ANCHOR both tack wires — epoxy/glue*
☐	**12.**	RXCAN tack wire (pad 4)	→	#2 CAN RX	
☐	**13.**	#2 CANH / CANL	→	bus screw terminal or pigtail	*the ONLY signal leaving Board A*

Board A stays inside the dash enclosure with the Pi. No 5V anywhere on Board A.

BOARD B — DAQ node (Teensy 4.1 + SN65HVD230 #1 + dividers + terminals)

☐	**14.**	Teensy 3.3V	→	#1 3.3V + B: 3V3 rail	
☐	**15.**	Teensy GND	→	B: GND rail	
☐	**16.**	Teensy pin 22 (CAN1 TX)	→	#1 CAN TX	
☐	**17.**	Teensy pin 23 (CAN1 RX)	→	#1 CAN RX	
☐	**18.**	Teensy VIN (5V from USB)	→	B: 5V SENDER rail	*senders + rail-ref only*
☐	**19.**	#1 CANH / CANL	→	bus screw terminal or pigtail	
☐	**20.**	Teensy USB	→	Pi USB	*bench: power+serial+reflash. Car: node gets a local 12->5V buck to VIN instead; reflash = laptop at the node*

Pins 22/23 belong to CAN — the analog bank uses 14-21 only.

THE BUS — Board A <-> Board B

☐	**21.**	A CANH	→	B CANH	*twisted pair; car run: twisted, away from ignition*
☐	**22.**	A CANL	→	B CANL	
☐	**23.**	A GND	→	B GND	*common ground (bench: via Pi USB anyway)*

TERMINATION: 120R at the two physical ENDS of the bus. Two boards = both terminate (as today).

When a THIRD node joins later, only the two end nodes keep 120R — disable the middle one's.

Two-board sheet — page 2: Board B dividers, terminals, checks

E. Divider bank on Board B — TERMINAL → 2.2k → node → 3.3k → GND, node → Teensy pin

ch	purpose	terminal	Teensy pin	firmware
1	Oil pressure (pot today)	T3	A0 / pin 14	live (PIN_OILP)
2	Fuel pressure	T4	A1 / pin 15	reserved
3	AFR (wideband analog)	T5	A2 / pin 16	reserved
4	MAP	T6	A3 / pin 17	reserved
5	Oil temp (thermistor)	T7	A4 / pin 18	reserved — see note
6	5V rail reference	(internal)	A5 / pin 19	reserved — ratiometric fix
-	spares	T9/T10	pins 20/21	unwired, bring to pads

Ch6 input comes FROM Board B's own 5V sender rail through an identical divider — no external wire.
Ch5 oil temp: thermistor needs a PULL-UP, not a divider — leave resistor spots empty until the AEM 30-2012 arrives and the curve build sets values. Terminal + pin routing stays the same.
Verify EVERY divider before power: input->node = 2.2k, node->GND = 3.3k.

F. Board B screw terminals (dev face) — future Deutsch DT pins reserved

term	signal	future DT pin
T1	+5V sender supply (fused later)	1
T2	sender GND	2
T3	oil pressure signal	3
T4	fuel pressure signal	4
T5	AFR signal	5
T6	MAP signal	6
T7	oil temp signal	7
T8	sender GND (second)	8
T9	spare analog (divided!)	9
T10	spare digital 3.3V-ONLY	10
T11	turn-L tap (VIA PROTECTION, car phase)	11
T12	turn-R tap (VIA PROTECTION, car phase)	12

T11/T12 land on the strip now but stay UNWIRED to the Teensy until the 12V protection circuit exists.

G. Final checks — before first power-on

- ☐ Continuity: every Pi SPI line to its MCP2515 pin, ribbon seated/soldered solid
- ☐ 60 ohms across CANH-CANL with both boards connected, power off
- ☐ Every divider: 2.2k top / 3.3k bottom, verified with the meter per channel
- ☐ NO path from Board B's 5V rail to any Teensy pin except THROUGH a divider
- ☐ TXCAN/RXCAN tack wires anchored (epoxy/hot glue) with strain relief
- ☐ IMU flat + rigid on Board A, orientation noted here: ______
- ☐ USB, ribbon, and bus pigtails strain-relieved to their boards
- ☐ Power-on order: Pi alone -> candump shows Teensy frames -> pot sweep on T3 -> in-situ reflash test

Keep the breadboard alive until every check passes. Board B is the prototype of the DAQ-node PCB.